



Towards the resilience quantification of (military) unmanned ground vehicles

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ABSTRACT

In the case of Unmanned Ground Vehicles (UGVs), resilience can be an economical, an environmental, but most importantly, a mission-critical question as well: mission failure caused by the lack of resilience in some cases might imply the loss of the UGV, which could lead to human and financial losses and environmental damage. Thus, the aim of this article is to provide a methodology for UGV resilience analysis by introducing a generalizable method that can be applied both for complete UGV systems and subsystems, and leads to resilience quantification. After proposing a specific resilience definition for UGVs, this article proposes a method for UGV resilience assessment using process graphs, created based on the system components and the expected behavior of UGVs. To provide a context for the introduced solution, existing methods applied for UGV resilience assessment are briefly mentioned. The application of the proposed method is showcased on the perception subsystem of a UGV, finalized with the evaluation of the achieved results.

1. Introduction

During the recent years, the involvement of teleoperated, semi-autonomous and autonomous Unmanned Ground Vehicles (UGVs) has seen a major increase both in case of civil and military applications. In both cases, the main motivator behind UGV application is the goal to exempt human workforce for Dirty, Dull, Dangerous (3D) tasks. These use-cases imply the fact that UGVs are often partially or fully damaged during their missions, which might lead to parts, components, or complete UGV systems left at the location of the mission. A lost, non-functional UGV causes problems from three perspectives. Firstly, a lost UGV means a partially, or completely failed mission. Secondly, UGVs usually contain components that require special disposal processes, causing a significant environmental impact when abandoned. Discarded UGVs can be considered to be hazardous waste, as they contain batteries or fuel. Despite different directives that regulate the composition of electronics, sensors and computational devices are still considered to be hazardous waste. Thus, it is straightforward that the increasing number of UGV applications is making the matter of discarded, lost UGVs more and more serious. The possibility for unattended waste creation by discarding UGVs during missions can be decreased by making UGVs more resilient. Finally, it is also important to consider the fact that UGVs are low-quantity products, containing expensive, unique components manufactured using special materials. Thus, unexpected losses have to be assessed from an economic point of view as well. To enable the deployment of autonomous systems, it is necessary to integrate the

survivability paradigm as one of their primary capabilities. Such systems are required to exhibit minimal human intervention, long endurance, task persistence, and the ability to surmount unexpected hazards and situations, as these are key traits for achieving autonomy (Quek et al., 2006). Decreasing the number of failed critical UGV missions and decreasing unattended waste creation by discarding UGVs during missions can be simultaneously achieved by making UGVs more resilient.

Current approaches of resilience-oriented design often consider steps which add redundancies to the system, increasing item costs as well. The resilience of UGVs can be approached from many perspectives, since UGVs can be described as special vehicles or complex computational systems with actuators. Currently, there is no widely accepted definition for the resilience of UGVs or autonomous mobile robots. However, there is a general aspiration for designing resilient propulsion platforms, sensing systems and computational devices for such vehicles. Some design approaches considering resilience and survivability in UGVs are resilience-by-design and resilience-by-reaction, aimed at designing cyber systems with resilience in mind and adapting the system to failures, respectively (Bagchi et al., 2020). Additionally, human-machine semi-autonomous systems can rely on human decision-making for critical situations, with control systems designed to overcome unexpected disturbances (Zieba et al., 2011). Graceful degradation has been proposed in order to maintain the UGV's partial functionality in case of failures without significantly increasing the complexity of the software (Adam et al., 2016). A different approach considered the autonomous vehicle as a complex system, and assessed its resilience using a graph

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method (Bucić et al., 2018). From a solely military perspective, UGVs can be seen as components of a mission, and the success of the mission depends on the resilience of this system (Sullivan et al., 2018). In addition to general resilience assessment methods, some approaches solely focus on increasing the resilience of given platforms or systems (Hernández et al., 2018). To create a resilience model for semi-autonomous or autonomous unmanned ground vehicles (UGVs), a comprehensive analysis of the main hardware and software components is required. Subsequently, the UGV should be evaluated as a system. Thus, the recommended inputs for resilience assessment encompass an evaluation of the resilience of individual components, such as sensors, telecommunication devices, high-level and low-level computers, and actuators. Additionally, the level of modularity of the system, redundancy of the components, grade of software fault handling, and physical protection of hardware components should be considered (Hajdu et al., 2022).

1.1. Current approaches for UGV resilience assessment

Despite there are no widely accepted methods for UGV resilience assessment, some approaches can be found. The following approaches are mentioned for the purpose of context, but are not applied in this paper.

1.1.1. Graph-based approach

The resilience problem of autonomous vehicles and mobile robots is usually addressed as a separate design requirement for each given robot design, there is no widely accepted model or approach. There are however some approaches which are aiming to provide a general model and method for UGV resilience assessment and quantification. Bucić et al. (2018) presents an initial discussion on the control, design, and motion planning abilities of an autonomous system in which the controller experiences a partial loss of control authority. The primary focus of the paper is to develop sufficient and necessary conditions for the existence of a safe control policy in such a partially controlled system. To obtain these conditions, the system's motion was interpreted as a type of adversarial safety game on a graph, where one player's moves include labelling the edges of the game graph. The paper demonstrates that the safety objective in the original control system can only be achieved if such a game has a winning strategy, which occurs if and only if there exists a labelling of the game graph that satisfies specific properties. The authors found a sufficient and a necessary condition for the existence of such a labelling in terms of minimal degrees of a subgraph of the original graph and discussed how those conditions apply to the motion of an autonomous vehicle and to communication using a set of codewords of length n with a bounded running digital sum. The future work is to broaden the scope of the considered framework, including considering a broader class of control specifications, such as temporal logic specifications, and developing graph-based criteria for the existence of a system design that allows for a correct control policy. This work can be considered to be a significant step towards ensuring system resilience under partial loss of control authority. Bucić et al. (2018) formulate the problem as a discrete-time system with dynamics given by the following equation:

$$x(t+1) = x(t) + u(t) \quad (1)$$

The system is controlled by actuators $A_1, A_2, \dots, A_p$, and the control effort is given as $u(a_1(t), \dots, a_p(t))$, where $a_i(t) \in A_i$. The control objective is safety, which means that the system should be in a predetermined set $S \subseteq Z_n$ for all $t \geq 0$. The article poses two questions:

- Is it possible to determine a control policy that guarantees system safety for all possible choices of $a_1(t), \dots, a_r(t)$?
- Is it possible to design sets $U(a_1, \dots, a_r)$ so that the control policy formulated in the previous question exists?

Furthermore, the article formulates the Restricted Partition Control Problem (RPCP) and the Free Partition Control Problem (FPCP) to address the questions stated before. The RPCP asks whether there exists a function $\hat{u} : U_{i=1}^\infty = [m]^i \rightarrow \mathbb{R}$ that satisfies two conditions:

- $\hat{u}(d_1, \dots, d_k) \in U(d_k)$ for all $d_1, \dots, d_k \in [m]$
- for every $d : \mathbb{N}_0 \rightarrow [m]$, if $x(t)$ is the solution of (1) with $u(t) = \hat{u}(d(0), \dots, d(t))$, then $x(t) \in S$ for all $t \in \mathbb{N}_0$. The FPCP asks whether there exists a partition $\{U(1), \dots, U(m)\}$ such that the RPCP admits a solution.

The proposed method by Bucić et al. (2018) continues by game formulation, a step which formulates the RPCP as a winning strategy in case of a two-player game. This step simplifies the initial problem into a graph labelling task.

This method approaches resilience from the point of view of the vehicle's ability to perform specific tasks.

1.1.2. Fault-tolerant system design

Fault-tolerant UGV design is the process of creating UGVs that can continue to operate even in the presence of component or system failures. This design approach involves identifying potential faults or failures and implementing mechanisms to detect, isolate, and recover from them, this being the main difference between fault-tolerant design and general UGV design with resilience in mind. Fault-tolerant design lists possible failures and proposes design methods to prevent these possible failures, while resilient design aims to make the UGV non-dependent of any, unexpected failures.

Fault-tolerant UGV design typically involves redundancy in critical components, such as sensors, actuators, and computing systems. This redundancy allows the vehicle to continue operating even if one or more components fail. Additionally, fault-tolerant UGV design often includes mechanisms for fault detection and diagnosis, which can help identify and isolate faults before they cause serious problems. Other design considerations for fault-tolerant UGVs include robustness to environmental factors, such as dust, moisture, and temperature extremes, as well as the ability to adapt to changing mission requirements.

In contrast to fault-tolerant systems, resilient systems are designed to adapt to changing conditions and recover quickly from minor failures that do not require repair, whether they are caused by component failures, environmental factors, or other types of events. Resilient systems are often characterized by their ability to neglect different kinds of disturbances and maintain their core functions even after or during significant challenges. Resilient design typically involves redundancy, modularity, and flexibility, as well as the ability to detect and respond to changes in the system's environment or operating conditions.

1.1.3. Unsupervised learning algorithm

Ma et al. (2020) defined this theory primarily for surface vehicles (USV). This contribution has developed an unsupervised learning strategy, where the algorithm needed to propose different tasks for a multi-USV system and then with an application of a K-means algorithm a self-organizing map has been created and modelled how to deal with the execution. These problems are identified and assessed. The novelty of this methodology was the assignment of a new competition strategy for algorithm.

It is important to note that UGVs and USVs are not operated in the same area therefore these methods application should be carefully reviewed; however, this unsupervised learning methodology could be also a guiding principle.

2. Method

In order to create a method for UGV resilience quantification, it was necessary to define the possible system states. The general definition of

resilience allows different constraints and allowances when considering UGV resilience. The following questions had to be addressed:

- Are system-level or component-level, partial or total fixes allowed after a fault?
- Is the UGV considered to be functional if:
 - o It is able to fully maintain its capabilities, thus, able to fulfil missions.
 - o It is able to maintain its autonomous locomotion capability.
 - o It is able to partially maintain its autonomous capability in order to carry out emergency missions (return home).

Regarding system and component reparability, we have considered theoretic mission formulations and real-life examples. Based on [Carlson and Murphy \(2005\)](#), 80% of UGV failures are terminal, which means that in these cases the UGV is not able to continue its mission. It is important to mention that this value was determined based on case studies and field reports using teleoperated, non-autonomous UGVs. However, the main physical components are similar in case of teleoperated and autonomous UGVs, only the on-board software being significantly different, this result is applicable without compromising the final decision. As we are considering short missions with high risks using autonomous UGVs, a terminal failure leaves no room for repairs, and it is very likely that it leads to the loss of the UGV.

Based on the approach described above, we have introduced an evaluation system addressing the autonomous capability of the UGV. When considering the way of resilience determination, we focused on the high-level control system. We were mainly focusing on the effects of component redundancies and sensor fusions. It is important to highlight that in case of autonomous systems, they do not necessarily lose their autonomous abilities completely. We have introduced q for quantifying autonomous ability. As it can be observed in [Table 1](#), the value of q defines given levels of capability, and based on this scoring, the system or subsystem completely loses its ability if q drops below 0.2. The quantification of the capability levels is presented in [Table 1](#).

The terms used in [Table 1](#) were determined based on the behavior of subsystems of autonomous vehicles in case of failures with different severity. The presented quantification is a convention provided by us to express system capability levels with values normalized between 0 and 1. The mentioned levels of autonomous capabilities can be described as follows:

- Loss of capability: the analyzed system or subsystem is unable to carry out its functionality, it is unable to provide any useable output.
- Emergency functions: the analyzed system or subsystem is unable to provide useable output, but it is able to detect its malfunctions and carry out emergency protocols that allow other dependent system components to maintain their functionality.
- Minimal capability: analyzed system or subsystem is able to carry out a minor portion of its functionality, which might be enough to maintain the locomotion capability of the UGV.
- Compromised capability: the analyzed system or subsystem is able to fulfil its designed functionality with minor errors inaccuracies.
- Full capability: the analyzed system or subsystem is able to perform its intended functionalities.

Table 1
The quantified capability levels of systems of UGVs.

Level of capability	Description
$q < 0.2$	Loss of capability
$0.2 \leq q < 0.6$	Emergency functions
$0.6 \leq q < 0.8$	Minimal capability
$0.8 \leq q < 1$	Compromised capability
1	Full capability

It is important to mention that the proposed capability quantification can be carried out based on the in-depth understanding of the assessed UGV.

Our proposed method is structure construction based on P-Graphs. A P-Graph, short for Process Graph, is a distinctive bipartite graph that illustrates the structure of a process system. The graph displays operating units as horizontal bars, and their input and output materials as solid circles. This graph is directed, with the arcs indicating the direction of material flows in the network. Specifically, the arcs are directed to an operating unit from its input materials and from an operating unit to its output materials ([Friedler et al., 1992a,b](#)).

The resilience of processing systems was initially formulated by [Orosz et al. \(2020\)](#), and this paper presents a more generalized formulation of the concept. The initial formulation is focused on the binary case, where processing systems have two states: fully operational and non-operational (total failure). Later, this idea was expanded on this by considering more complex cases where operating units may exhibit various levels of partial operations in addition to the fully operational and total failure states ([Orosz et al., 2021](#)). Let us consider a processing system that produces a single product. β represents the normal production flowrate, while γ denotes the minimum acceptable production flowrate in terms of the system's resilience, and δ represents the minimum production flowrate technically achievable by the system. Resilience of a system, as defined in available literature, refers to the system's ability to transit from non-normal operations to normal operation. In practice, however, knowledge about the system's capability to return to a pre-specified production level may provide valuable information about the system's behavior. For instance, if the system can achieve 80% of the production level during critical conditions, it is essential to understand the system's resilience for this level of production. γ denotes this acceptable reduced production level, and its value must be defined by the designer based on their previous knowledge of the system and specific requirements. If the designer does not specify γ , its value is assumed to be equal to β . The parameter γ provides the designer with an opportunity to reveal additional information about the system's behavior in terms of resilience. Under normal circumstances, any part of the processing system may fail, and it is therefore important to assess the system's resilience in the face of such failures. Formally, if the processing system is operating normally and an unexpected event occurs, causing some or all of the system's operating units to be partially or completely destroyed, it is difficult to determine or predict the resulting effects and consequences due to the unexpected nature of the event. As a result, all possible combinations of failures must be considered in order to determine the system's resilience.

The general formulation of resilience of production systems considers system components as reparable components, calculating with component-specific repairation times. This approach has to be modified when considering on-board systems of UGVs without the possibility of component replacement or repair.

[Orosz et al. \(2022\)](#) describes the general methodology as follows: For any given natural number n , the set S_n^f represents all the possible combinations of up to n simultaneous failures that a processing system can experience during normal operation at time 0. Additionally, the cardinality of S_n^f is denoted as N_n^f , which is the number of elements in the set S_n^f . If the processing system consists of K operating units, then the set of all possible failure combinations is denoted as $S^f = S_K^f$. The cardinality of S^f is represented as N^f .

Furthermore, an element s from S_n^f (i.e., $s \in S_n^f$) represents a combination of failed and operational units. This element s is formally defined as a binary vector of size K , where each s_i ($i = 1, 2, \dots, K$) indicates whether the corresponding unit i is operational ($s_i = 1$) or not ($s_i = 0$). Thus, ($s_i = 1$) if the unit is operational, and ($s_i = 0$) if the given unit fails. This method can be implemented accordingly in case of UGVs as well. However, since we have considered that component or system level

repair is not possible in case of UGVs, further modelling has to be considered discrete, without repair times. Resilience can be determined by determining the states that maintain autonomy and states that cannot ensure that the UGV stays autonomous.

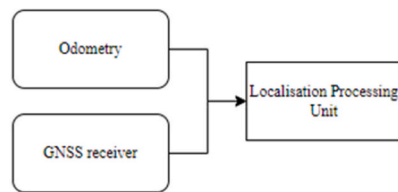
Based on our proposal, in case of UGV resilience assessment, P-Graph construction follows the logic of the required outcomes regarding the expected capability of the system. Based on our approach, the expected system state outcomes (called Product in P-Graph terminology) are in accordance with the possible system capabilities. Then, the inputs of the graph can be defined based on the inputs of the given system or subsystem. Connections between the available inputs and the expected output behavior can be created based on the data flow of the system or subsystem. The method can be applied both for UGV subsystems, such as environmental perception subsystem, propulsion subsystem, etc., or for the entire UGV system. In summary, P-Graph creation based on a UGV's on-board system and expected behavior can be carried out by following these steps:

- Determining expectable system/subsystem behaviors in case of different fault states. These expectable behaviors will form the Products in case of the P-Graph.
- Listing the inputs of the system/subsystem. As an example, environmental sensors provide input for the environmental perception subsystem, and high-level control provides input for the low-level control subsystem.
- Determining capabilities that are results of the proper functioning of distinctive groups of system components. These internal capabilities lead to the overall functioning of the system/subsystem. The results of these capabilities are called conditions from the point of view of the final expected behavior. These conditions can be represented as intermediate information of the P-Graph.

An illustrative example can be seen in Fig. 1. The UGV's localization module relies on two sources of input, namely the GNSS receiver and the odometry, one being an exteroceptive sensor and the other one information based on proprioceptive data. The GNSS receiver and the module providing odometry data appear on the P-Graph as operating units. Their outputs can be illustrated as intermediate information, namely Positioning and Location. Then, a virtual component has to be introduced to symbolize the fact that both position and location are required for a properly functioning localization. The added virtual component cannot enter into a faulty state. In case of this example, the final output is the non-compromised, available location information.

After the creation of the complete P-Graph, the resilience of the UGV can be evaluated based on the total number of autonomous and non-autonomous system states. This calculation can be carried out by using the SSG Algorithm (Friedler et al., 1992a,b).

After the P-Graph has been set up, the resilience index of the UGV can be determined based on (1). The resilience index is a normalized value ($0 \leq R \leq 1$), q is the level of capability of the UGV or one of its subsystems, and M is the number of total components of the system.



$$R(q) = \frac{C(q)}{2^{|M|}}, C \in |M|, M \in \mathbb{N}^*, 0 \leq R \leq 1 \quad (2)$$

3. Results and discussion

In order to carry out a quantifiable resilience assessment, a UGV system model is required. A complete, general UGV model is used, including environmental sensors, on-board computers, and actuators. By starting from a complete model, it is possible to assess separate sub-segments from the point of view of resilience.

This general block diagram of an UGV features all the usual functions required for autonomous locomotion, such as environmental perception, high-level control, low-level control and differential-drive self-propulsion. In case of environmental perception, it has two additional computational modules, one being a machine vision unit for depth camera perception and the other one being a separate computational module for localization based on a GNSS receiver. To provide a better understanding of the UGV's components, it is important to assess the system on a component level.

- **Odometry** - Odometry is the use of data from motion sensors to estimate the position and orientation of a mobile robot. It is a process of tracking the movement of a robot by measuring its wheel rotations, velocities, and accelerations. The goal of odometry is to estimate the robot's position and orientation over time, as it moves through an environment. Odometry works by using sensors such as wheel encoders, gyroscopes, accelerometers, and other motion sensors to measure the robot's movement. Wheel encoders measure the rotation of the robot's wheels, while gyroscopes and accelerometers measure its angular and linear velocities and accelerations. By combining this data, odometry algorithms can estimate the robot's position and orientation. Odometry is a widely used technique in mobile robotics, and it is often used in conjunction with other sensors such as cameras, lidars, and GPS to provide more accurate estimates of a robot's position and orientation. However, it is important to note that odometry estimates can be subject to errors due to wheel slip, sensor noise, and other factors, which can accumulate over time and lead to drift in the robot's estimated position. Therefore, odometry should be used as part of a larger system that can correct for these errors and provide more accurate estimates of a robot's pose.
- **GNSS receiver** - GNSS, or Global Navigation Satellite System, is a technology used for determining the precise location of an object on the Earth's surface. GNSS is a system of satellites that transmit signals to a receiver on the ground, which uses those signals to calculate its position. The most commonly used GNSS is the Global Positioning System (GPS), but GNSS receivers can also receive signals from other satellite systems, such as the Russian GLONASS system, the European Galileo system, and the Chinese Beidou system. By combining signals from multiple GNSS systems, a receiver can improve its accuracy and reliability.
- **Lidar sensor** - Lidar, or Light Detection and Ranging, is a remote sensing technology that uses laser light to measure distances and

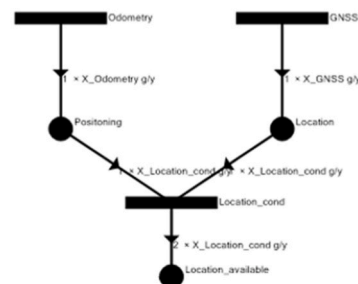


Fig. 1. Example P-Graph creation based on block diagram.

create three-dimensional maps of the environment. Lidar works by emitting short pulses of laser light, which bounce off of objects in the environment and return to the sensor, where they are used to create a detailed 3D map of the scene. Lidar sensors consist of a laser, a scanner, and a receiver. The laser emits short pulses of light in a specific wavelength, typically in the near-infrared range. The scanner directs the laser beam to different points in the environment, and the receiver measures the time it takes for the laser pulses to bounce back from objects in the scene. By measuring the time delay between the emitted laser pulse and the received signal, lidar can determine the distance to objects in the scene with high precision. Lidar can also measure the intensity of the returned signal, which can be used to distinguish between different types of objects in the scene.

- **Color camera** - A color camera for mobile robots is a device that uses visible light to capture images of the robot's environment. Color cameras for mobile robots typically consist of a lens, an image sensor, and supporting electronics. The lens focuses the light from the environment onto the image sensor, which converts the light into electrical signals that are processed to create an image. The supporting electronics may include digital signal processing circuits to enhance image quality and reduce noise. Color cameras for mobile robots are used in a variety of applications, such as object recognition and visual navigation. They can provide rich visual information about the robot's environment, including details such as texture, color, and shape, which can be used to identify objects and features in the scene.
- **Depth camera** - about the robot's environment. Depth cameras use a variety of techniques to measure the distance to objects in the scene, such as structured light, time-of-flight, or stereo vision. Depth cameras can provide more detailed information about the environment than traditional color cameras, as they can capture 3D information about the scene. This information can be used for applications such as object recognition, navigation, and obstacle avoidance.
- **High-level controller** - High-level controllers can be used to plan and execute complex behaviors and tasks, such as navigation, exploration, mapping, and object manipulation. They can also integrate inputs from multiple sensors and coordinate the actions of multiple actuators, such as motors, grippers, or cameras. In case of mobile robots, the main task of the high-level controller is path planning. The expected path is sent to the low-level controller.
- **Localization processing unit** - The localization processing unit provides real-time position and orientation information based on data received from the GNSS and the odometry system.
- **Machine vision processing unit** - The machine vision processing unit is a specialized hardware component that provides high-performance processing and analysis of visual data from the robot's cameras and sensors. The MVPU is designed to handle the computationally intensive tasks required for vision processing, such as object recognition, tracking, and classification.
- **Low-level controller** - The low-level controller is responsible for implementing basic control functions such as motion control, sensor data processing, and data communication. These functions may include tasks such as motor speed control, sensor calibration, and data filtering and smoothing. It usually includes a real-time micro-controller, motor speed controllers, sensor data processing units and communication modules.
- **Propulsion** - A UGV's propulsion system contains the engines or motors and additional mechanisms (e.g., torque converters) that drive the UGV's wheels. These propulsion systems can apply internal combustion engines as well, but because of its mechanical simplicity, reduced operational noise and lower environmental impact, electric drive has become more popular. UGVs can have many drive configurations, the most popular configuration being the differential-drive, featuring at least two independently controllable drive axles on the two opposite sides of the robot platform.

To showcase the proposed resilience assessment method, sub-segments of the general block diagram have been transformed into a P-graph representation, and the subsegment responsible for environmental perception is further analyzed from the point of view of resilience.

Environmental perception is a key capability of autonomous UGVs. Since it is carried out by specific components, perception submodules can be easily identified. For the purpose of this use-case, a sample perception unit is considered based on the block diagram presented in Fig. 2. In this case, environmental perception is achieved by object localization and object classification. Object localization is carried out based on the input of a lidar sensor (3D point cloud) using a processing unit (PU). The result is object localization, which means that the UGV is aware of the presence and the relative location of an object, but cannot identify, classify it. Object localization is a partial result of the process, which can already be used for avoiding obstacles, but it can further improved by the addition of object classification. Object classification can be implemented by using a camera sensor and a graphical processing unit (GPU). The result of object classification is the basic recognition of objects out of specific object classes, e.g., labelling the surrounding vegetation, road surface, humans, vehicles etc. By combining the output (carried out by another processing unit) from object localization and object classification, a basic environmental perception can be achieved. The described process is represented by the P-graph presented in Fig. 3.

In this case, depending on the combination of component failures, the perception capability levels can be described as follows:

- no perception capability
- object localization
- object classification.

Numeric values have been assigned to these q capability levels normalized to $(0 \leq q \leq 1)$ interval. The values have been determined based on the applicability of the data output from the point of view of autonomous navigation. In this case, object detection allows complete functionality from the point of view of environmental perception, thus it receives value $q = 1$. In a case in which only object localization is available, simple obstacle avoidance can be carried out, a return home mission is possible, but full autonomy is compromised. This means the abruptness of the mission and reduced locomotion capabilities as well, thus, this case receives value $q = 0.4$. Finally, if there is no available perception capability, $q = 0$ is assigned. Provided that based on our convention, the level of perception capability allows some level of autonomous locomotion if $q \geq 0.2$, resilience from the point of view of perception can be determined by taking into consideration all the cases in which the autonomous capability is above this level. Based on the P-graph (Fig. 3), there number of cases in which the perception level is $q = 1$ is 1, because this capability level requires all components to be functional. In case of a capability level of $q = 0.4$, the substructure ensuring object localization has to be fully functional and any of the components can fail on the other branch, meaning $2^3 = 8$ cases. Finally, there are $2^5 - 2^3 - 1 = 23$ cases in which the level of perception capability is $q = 0$. The assignment and the case number of the mentioned values is presented by Table 2.

If a perception capability of $q \geq 0.2$ is required, based on (2), the resilience index can be calculated as follows:

$$R(q) = \frac{C(q)}{2^{|M|}} = \frac{1+8}{32} \cong 0.281 \quad (3)$$

It has to be underlined that the resulting resilience index is the characteristic of a system which was not designed with resilience in mind. Autonomous systems and subsystems such as environmental perception tend to rely on multiple sensor inputs and there are often separate, dedicated processing units for different tasks, without redundancy. Thus, it is common that the failure of a single module, such as a sensor or a processing unit leads to the loss of specific capabilities. From

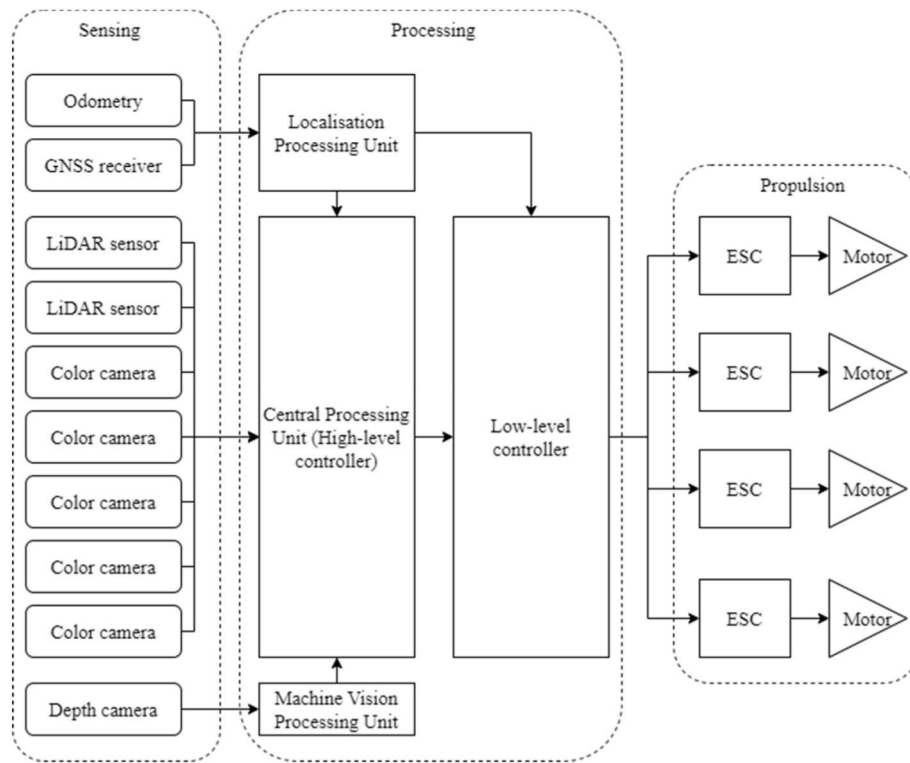


Fig. 2. Block diagram of a UGV's system components.

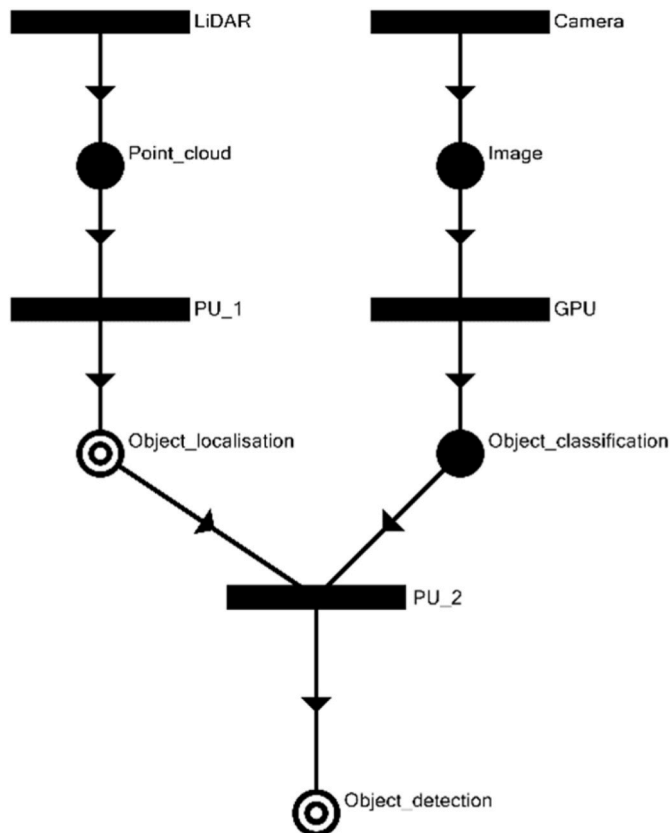


Fig. 3. P-graph representation of an autonomous UGV's perception unit.

Table 2

Quantified perception capability levels.

Number of cases	Level of perception capability	Description
$2^5 - 2^3 - 1 = 23$	0	No perception
$2^3 = 8$	0.4	Object localization, reduced perception
1	1	Object detection, full perception

this perspective, the resulting resilience index of $R(q) = 0.281$ is a reasonable result.

4. Conclusion and further research

Resilience assessment plays a crucial role from the perspective of unmanaged waste reduction caused by lost autonomous UGVs during missions. Despite there are multiple methods for resilience assessment, there is no widely accepted approach specifically for UGV resilience quantification. Therefore, this article presents a possible methodology for this purpose using P-graphs. The proposed methodology is applicable for entire UGV system models and for subsystems, such as perception units as well. It has been taken into consideration that some systems and subsystems might produce partial output results which might be enough for the reduced functioning of the UGV. The proposed method has been applied on a perception subsystem, and a resilience index for this subsystem has been determined.

One major limitation of the presented method is the fact that quantified system and subsystem capability levels have to be determined based on the in-depth understanding of the given system. This requires rigorous component-level analysis in case of the assessed UGVs.

Future research scopes include the validation and further enhancement of the proposed solution. Validation is planned to be carried out using models of real UGV systems. Furthermore, it is a future scope to create an additional methodology that assists the quantification of system and subsystem capability levels. This additional method would

significantly decrease the current limitation of the presented method.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- Adam, S., Kuhrmann, M., Schultz, U.P., 2016. Towards a virtual machine approach to resilient and safe mobile robots. In: 2016 IEEE 21st International Conference on Emerging Technologies and Factory Automation (ETFA). Presented at the 2016 IEEE 21st International Conference on Emerging Technologies and Factory Automation (ETFA). IEEE, Berlin, Germany, pp. 1–8. <https://doi.org/10.1109/ETFA.2016.7733545>.
- Bagchi, S., Aggarwal, V., Chaterji, S., Douglass, F., El Gamal, A., Han, J., Henz, B., Hoffmann, H., Jana, S., Kulkarni, M., Lin, F., Marais, K., Mittal, P., Mou, S., Qiu, X., Scutari, G., 2020. Vision paper: grand challenges in resilience: autonomous system resilience through design and runtime measures. IEEE OPEN JOURNAL OF THE COMPUTER SOCIETY 1, 155–172. <https://doi.org/10.1109/OJCS.2020.3006807>.
- Bucić, M., Ornik, M., Topcu, U., 2018. Graph-based controller synthesis for safety-constrained, resilient systems. In: 2018 56th Annual Allerton Conference on Communication, Control, and Computing (Allerton). Presented at the 2018 56th Annual Allerton Conference on Communication, Control, and Computing (Allerton). IEEE, Monticello, IL, USA, pp. 297–304. <https://doi.org/10.1109/ALLERTON.2018.8635905>.
- Carlson, J., Murphy, R.R., 2005. How UGVs physically fail in the field. IEEE Trans. Robot. 21, 423–437. <https://doi.org/10.1109/TRO.2004.838027>.
- Friedler, F., Tarján, K., Huang, Y.W., Fan, L.T., 1992a. Graph-theoretic approach to process synthesis: axioms and theorems. Chem. Eng. Sci. 47, 1973–1988. [https://doi.org/10.1016/0009-2509\(92\)80315-4](https://doi.org/10.1016/0009-2509(92)80315-4).
- Friedler, F., Tarján, K., Huang, Y.W., Fan, L.T., 1992b. Combinatorial algorithms for process synthesis. Comput. Chem. Eng. 16 [https://doi.org/10.1016/S0098-1354\(09\)80037-9](https://doi.org/10.1016/S0098-1354(09)80037-9).
- Hajdu, A., Krecht, R., Suta, A., Tóth, Á., Friedler, F., 2022. The resilience barriers of automated ground vehicles from military perspectives. Chemical Engineering Transactions 94, 1195–1200. <https://doi.org/10.3303/CET2294199>.
- Hernández, C., Bermejo-Alonso, J., Sanz, R., 2018. A self-adaptation framework based on functional knowledge for augmented autonomy in robots. ICA 25, 157–172. <https://doi.org/10.3233/ICA-180565>.
- Ma, S., Guo, W., Song, R., Liu, Y., 2020. Unsupervised learning based coordinated multi-task allocation for unmanned surface vehicles. Neurocomputing 420. <https://doi.org/10.1016/j.neucom.2020.09.031>.
- Orosz, Á., Pimentel, J., Argoti, A., Friedler, F., 2022. General formulation of resilience for designing process networks. Comput. Chem. Eng. 165 <https://doi.org/10.1016/j.compchemeng.2022.107932>.
- Orosz, Á., Pimentel, J., Friedler, F., 2021. Processing systems design considering resilience. 31st European symposium on computer aided process engineering. Computer Aided Chemical Engineering 50, 807–812. <https://doi.org/10.1016/B978-0-323-88506-5.50126-1>.
- Orosz, Á., Pimentel, J., Friedler, F., 2020. General formulation for the resilience of processing systems. Chemical Engineering Transactions 81, 859–864.
- Quek, B.-K., Ibañez-Guzmán, J., Lim, K.-W., 2006. Attaining operational survivability in an autonomous unmanned ground surveillance vehicle. In: IECON Proc. Presented at the IECON 2006 - 32nd Annual Conference on. IEEE Industrial Electronics, Paris, pp. 3969–3974. <https://doi.org/10.1109/IECON.2006.348001>.
- Sullivan, D., Colbert, E., Cowley, J., 2018. Mission resilience for future army tactical networks. In: 2018 Resilience Week (RWS). Presented at the 2018 Resilience Week (RWS). IEEE, Denver, CO, pp. 11–14. <https://doi.org/10.1109/RWEEK.2018.8473522>.
- Zieba, S., Polet, P., Vanderhaegen, F., 2011. Using adjustable autonomy and human-machine cooperation to make a human-machine system resilient - application to a ground robotic system. Inf. Sci. 181, 379–397. <https://doi.org/10.1016/j.ins.2010.09.035>.